PakEmis Sector Technical Note TN06

An Infrastructure-Constrained, Observation-Evaluated Inventory of Oil and Gas Fugitive Emissions in Pakistan

Rehan Ahmad[a,*]

[a] *Pakistan Air Quality Initiative (PAQI), Karachi, Pakistan*

Abstract

Fugitive emissions arise across segments that national production statistics alone cannot represent. We develop an infrastructure-constrained inventory for Pakistan from the 2006 IPCC Guidelines and the segment-resolved 2019 Refinement, adding every segment public data support. Gas balances, Pakistan's Common Reporting Table (CRT), flare records and methane-plume detections provide separate evaluation layers.

On common FY2023–24 production support, legacy and refined methods give 3.317 and 4.485 Mt CO_2e, respectively. The admitted public-data ledger gives an *upper-practice conservative (default-practice) configuration* of 23.499 Mt—not an empirically centred best estimate: segment coverage matters more than the factor revision. Joint structural configurations span 9.601 Mt to 24.694 Mt; the reference sits high in that envelope, and no probability attaches to any point in it. Uncertainty is dominated by gas-distribution practice and a transferred post-meter factor.

Adjacent-fiscal-year reconstructions bracket calendar 2022 at 23.761 Mt to 25.187 Mt against 4.749 Mt in Pakistan's 2022 CRT on a common GWP basis. The CRT arithmetic reproduces exactly, so the difference lies in activity, factor and boundary choices. Over the FY2015–16 to FY2023–24 span analysed here, declining upstream emissions are partly offset by gas-network growth. Matched detections occur near fields with 71.1 % of reported supply, yet production is only weakly rank-associated with observed activity.

Distribution technology and post-meter leakage are the two measurement priorities; the national release extends the module through FY2024–25 with annual source controls.

Keywords: fugitive emissions, methane, oil and gas, gas distribution, IPCC 2019 Refinement, satellite flaring, methane plumes, structural uncertainty, national inventory, Pakistan

PakEmis Sector Technical Note TN06. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

*Corresponding author
Email address: `rehan@pakairquality.com` (Rehan Ahmad)

Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

1. Introduction

Methane emissions from oil and natural-gas systems do not scale with production alone. They arise from wells, tanks, compressors, processing plants, pipelines, distribution networks, appliances and episodic releases; their magnitude

depends on equipment and operating practice. An inventory built only from national production therefore misses much of the source population and obscures the infrastructure where mitigation would occur.

The distinction is methodological. The 2006 IPCC Guidelines published broad Tier 1 ranges for developing-country systems, including ranges spanning orders of magnitude (IPCC, 2006). The 2019 Refinement introduced separate factors for oil production, transport and refining and for gas production, gathering, processing, transmission, distribution and post-meter sources. Several factors depend explicitly on leak-detection-and-repair (LDAR), pipeline material and compressor-seal practice (IPCC, 2019). Applying the Refinement in a data-scarce country therefore requires both a method change and a reconstruction of the physical supply chain.

Pakistan is a useful case. National production statistics are public, and the regulator reports utility throughput, pipeline length, compressor infrastructure, customers, LNG terminals, refinery production and unaccounted-for gas (UFG). Satellite-derived flare locations and methane-plume detections provide independent spatial evidence. Pakistan's official reporting, however, remains difficult to reconcile with these activity data, and direct national leakage measurements are unavailable.

This study addresses three questions. First, how much of the difference between legacy and refined fugitive estimates results from revised factors, and how much from adding previously omitted supply-chain segments? Second, which unresolved infrastructure characteristics dominate the national range? Third, what constraints are provided by utility balances, official reporting and satellite observations?

We answer these questions with a conserved segment inventory and physical gas-flow account, a method-resolution experiment, separate structural and parametric uncertainty analyses, and matched official and observational evaluations. The same source-neutral ledger can accept aggregate public controls today and facility-level records when they become available. Figure 1 summarizes the workflow.

2. Methodological context

2.1. From broad ranges to segment and practice factors

The 2006 method was defensible when production was the only widely available activity, but its developing-country ranges combine differences in source coverage, technology and practice. A geometric midpoint can represent the centre of a bounded log interval; it cannot identify which physical source is responsible for the result.

The 2019 Refinement changes that. For onshore gas production, for example, the methane factor is 4.09 tonnes CH_4 per million m^3 for higher-emitting practices and 2.54 for lower-emitting practices, both with reported uncertainty of ±20 %. Distribution can be calculated from gas consumption or pipeline length and is divided by plastic-pipeline and LDAR conditions. Transmission and processing similarly distinguish LDAR and compressor-seal practice. Where these characteristics are unknown, the guidance supports the higher-emitting/default selection and requires compilers to document it (IPCC, 2019).

2.2. Observation and inventory answer different questions

Bottom-up and atmospheric estimates often disagree because average factors omit abnormal or intermittent sources and because measurements sample different source populations. Aircraft and facility synthesis in the United States found supply-chain methane materially above the national inventory (Alvarez et al., 2018); other work showed that a small, heavy-tailed source population can reconcile apparently divergent estimates (Zavala-Araiza et al., 2015). Large sources also vary in time, so persistence cannot be inferred from a single overpass (Cusworth et al., 2021).

Satellite observations have made these releases independently visible. TROPOMI provides regional methane sensitivity (Veefkind et al., 2012); global analyses identify ultra-emitters (Lauvaux et al., 2022) and national fossil-fuel corrections (Chen et al., 2023). Such detections do not automatically produce annual national emissions: coverage, detection thresholds, intermittency and attribution still matter. They can, however, test whether an inventory's facility geography and source ranking are plausible. Global gridded fuel inventories face the complementary problem of spatially allocating national reports (Scarpelli et al., 2020, 2022).

The same distinction applies to flare observations. The World Bank's Global Gas Flaring Tracker derives annual flare volumes and individual flare locations from satellite radiance (World Bank, 2026). Because the 2019 production factors are inclusive of leakage, venting and flaring, adding observation-derived flare emissions to them would double count. Observed flaring is therefore retained as a memo and evaluation layer unless the inclusive factor is first disaggregated.

2.3. Official reporting as an audited comparator

Pakistan's Biennial Transparency Report and CRT provide the authoritative national account (Government of Pakistan, 2024, 2026). Authority does not make a comparison scope-matched automatically. The activity units, factors, notation keys, inventory year and GWP basis must be extracted and compared explicitly. We treat the official submission as a comparator whose derivation is to be reconstructed, not as a target to which the bottom-up model is forced.

3. Data and accounting boundaries

3.1. Reference-span infrastructure controls and the annual extension

Oil and gas production come from the Pakistan Economic Survey 2025–26, which reports 25.812 million barrels of crude oil and 1.141 million MMCF of natural gas in FY2023–24 (Finance Division, Government of Pakistan, 2026). Converted to

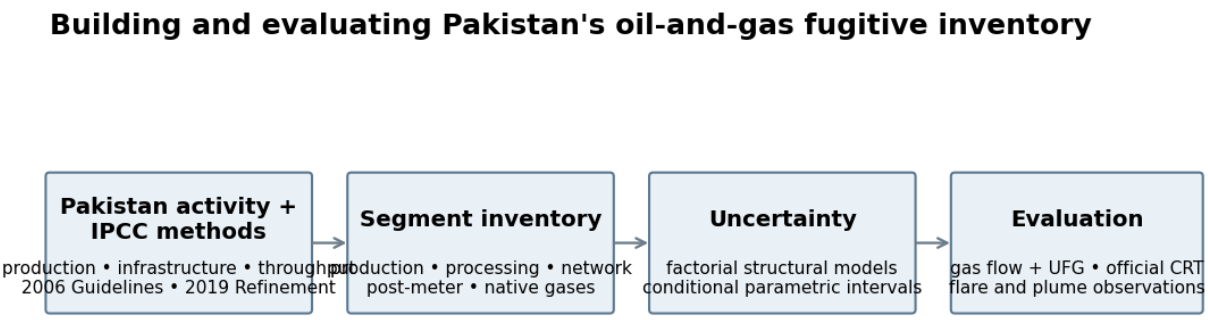


Figure 1: Scientific workflow from Pakistani activity data and IPCC methods through the segment inventory, uncertainty analysis and three independent evaluation layers.

IPCC activity units, these are 4103.77 thousand m^3 of oil and 32 309.47 million m^3 of gas.

The Oil and Gas Regulatory Authority (OGRA) archive supplies annual refinery, network, customer, consumption and UFG controls from FY2015–16 to FY2024–25 (Oil and Gas Regulatory Authority, 2026, 2023, 2024a,b). Values are transcribed from the reported annual tables with no interpolation. The image-based FY2020–21 report was independently OCR-audited against the page images, and the FY2016–17 refinery total is retained as the regulator's restated value. In FY2023–24, SNGPL reports 9320.69 km of transmission and 151 419 km of distribution pipeline; SSGCL reports 4206 and 49 811 km, respectively. The combined systems serve 10 748 538 domestic, 91 925 commercial and 11 318 industrial customers. Seventeen named compressor sites, two LNG import terminals and 3437.13 MMCF d^{-1} of utility and independent-system sales are reported. The reference T&D estimate uses measured pipeline length; throughput is an alternative IPCC basis. Across the FY2015–16 to FY2023–24 comparison span, transmission length rises from 11 555.08 km to 13 526.69 km, distribution length from 148 278 km to 201 230 km, and domestic customers from 7 997 096 to 10 748 538. These segment-resolved source records distinguish infrastructure changes from a production-only historical extrapolation.

The national core contains ten fiscal years, FY2015–16 through FY2024–25, with complete official annual controls for the appended year. The detailed endpoint quantities above describe FY2023–24; they are not relabelled as latest-year observations.

The FY2021–22 report identifies 8200 km of polythene distribution pipe in the SSGCL system. This is only 4.2 % of the reported national distribution network and does not classify the SNGPL network. Utility reports document active loss-control work—SNGPL surveyed 50 250 km of underground network in FY2023–24 and both utilities report rehabilitation and leak surveys—but do not establish national annual LDAR coverage (Sui Northern Gas Pipelines Limited, 2024; Sui Southern Gas Company Limited, 2024). Because the 2019 lower distribution factor requires both more than half plastic pipe and LDAR in use, the public evidence does not resolve either national criterion sufficiently to select it.

OGRA reports 10.67 million tonnes of refinery product output. IPCC calls for crude feed, so output converted at 0.85 t m^{-3} is a provisional activity proxy with 20 % uncertainty. Produced gas is similarly used as provisional gas-processing throughput. Neither proxy is hidden: both can be excluded by a coverage scenario.

Post-meter industrial activity is reconstructed directly from OGRA end-use controls rather than one undifferentiated non-residential total. The included FY2023–24 classes are commercial, general industry, fertilizer, captive power, cement and power stations, together 2439.23 MMCF d^{-1}, which converts to 25 211.03 million m^3 yr^{-1}; the registered activity the leak factor rides is 25 210.98 million m^3 yr^{-1} before the fertilizer feedstock is withheld and 18 684.62 after. The factor is defined on gas consumed in combustion equipment, and 83.5 % of the fertilizer class is chemical feedstock, reformed and not burned, on the producers' metered feed and fuel gas (National Fertilizer Development Centre, 2025); that feedstock, 6526.4 million m^3, is outside the base and the process fuel stays in. Residential gas is represented by the separate appliance fallback; CNG and unclassified other-system gas remain outside the industrial/power row. The included class sum differs from the registered six-class activity by only 0.042 million m^3 because published daily flows are rounded.

3.2. *Observational and official comparison layers*

The current World Bank individual-flare workbook (World Bank, 2026)—built on VIIRS Nightfire satellite pyrometry (Elvidge et al., 2016)—contains annual volumes for 2012–2025 and 130 sites labelled Pakistan. The UNEP MARS snapshot contains plume and source tables under a CC BY-NC-SA licence (United Nations Environment

Segment	Activity	Value	Status	Principal structural issue
Oil production	onshore oil	4,103.77 thousand m^3	official	offshore split unavailable
Oil refining	converted product output	12,552.94 thousand m^3	proxy	crude-feed throughput unavailable
Gas production/gathering	produced gas	32,309.47 million m^3	official/proxy	well and field split unavailable
Gas processing	produced gas	32,309.47 million m^3	proxy	plant throughput and sour share
Transmission	pipeline length	13,526.69 km	official	LDAR and seal practice
Distribution	pipeline length	201,230 km	official	material share and LDAR
Post-meter appliances	domestic connections	10,748,538	official/proxy	one appliance per meter
Post-meter industry/power	non-residential gas	25,210.98 million m^3	official	factor transferability; fertilizer feedstock withheld from this base (see text)

Table 1: FY2023–24 reference activities. A proxy is estimated with an explicit fallback and structural exclusion case; it is not described as observed.

Programme, 2026); after filtering country and sector, it contains 20 oil-and-gas sources and 174 plume detections. A public-domain geoBoundaries ADM1 layer assigns both products to provinces (Runfola et al., 2020). Field-level evaluation uses OGRA Table 3.34: its 84 published rows reconcile to 2637.75 MMCF d^{-1}, within 0.03 MMCF d^{-1} of the official total, and are canonicalized to 73 named fields.

The official comparison is extracted programmatically from Pakistan's 2022 CRT workbook. The pipeline stores the package and workbook SHA-256 hashes, worksheet names, cell coordinates, raw values, units, formulas and notation keys. The BTR's 2021 1.B.2 value is retained as a second official-year comparator.

3.3. Boundary and completeness

The reference account covers oil and natural-gas systems under 1.B.2. Coal mining fugitives (1.B.1) are outside the paper's stated boundary and are retained as NE in the completeness matrix, now with a partial Tier-1 screen beside the key: the Thar mine-mouth deliveries the power ledger holds bound the surface share (see `outputs/fugitives_coal_screen_2024.csv`), while underground production—whose per-tonne factors run thirty times the surface low—remains unquantified. Oil and gas exploration, abandoned wells, natural-gas vehicles and anomalous events also remain NE; storage is NO because the regulator describes facilities as planned rather than operating. Venting and flaring are included elsewhere (IE) in the inclusive segment factors and reported separately only as observation memos. For every NE row, the completeness matrix records whether it is potentially material, probably minor, unknown or outside scope and identifies the activity needed to estimate it. No numerical upper bound is invented where public activity is unavailable.

4. Inventory construction

4.1. Segment ledger and practice selection

For segment s and gas g, emissions are

$$E_{s,g} = A_s EF_{s,g}. \quad (1)$$

Each ledger row carries the IPCC code, activity and unit, source and imputation status, factor, uncertainty bounds and structural configuration. The accounting engine consumes this generic schema, allowing facility records to replace aggregate proxies without changing the downstream calculations.

The *reference configuration* uses the higher/default 2019 practice factors where national LDAR, pipeline-material or compressor-seal evidence is unavailable. Lower-practice factors and throughput-based transmission and distribution are evaluated as structural alternatives. Produced gas provides the processing proxy; sour-gas processing uses the 2019 default 32 % share. Both assumptions can be varied or excluded independently.

Post-meter appliances use one appliance per domestic meter and the Refinement's warm-country fallback. The industrial/power factor is 6.1 t CH_4 per million m^3, derived in the 2019 Refinement from a Revised 1996 value of 175 000 kg PJ^{-1} and a 35 MJ m^{-3} gas conversion. Its older empirical basis motivates full, half-factor and exclusion configurations. The complete factor lineage and uncertainty register are given in the supplement.

4.2. Method-resolution experiment

Five estimates separate factor changes from boundary expansion:

1. the legacy 2006 production-only calculation for FY2019–20;
2. the same method on FY2023–24 production;
3. 2019 factors on identical FY2023–24 oil and gas production;
4. the 2019 upstream account after adding gathering and processing; and
5. the public-data 2019 ledger, including T&D, LNG, refining and post-meter sources.

The common-support comparison isolates the factor-framework effect; the later steps measure the effect of representing additional physical segments.

4.3. Monthly timing, upstream only

The national release resolves this sector into fiscal months with process-specific support. Validated PBS provincial natural-gas and crude-oil production can support corresponding upstream production, gathering and processing timing as a throughput proxy. Twelve rows alone do not establish a complete national observation: source province

coverage, units and compatible monthly or annual controls are checked against the final exported panel. Repaired extraction and worksheet selection are reflected in the current profiles. Where the source remains incomplete, an explicit fallback replaces the broken profile and carries weaker evidence; it is not labelled national observed activity.

Distribution, transmission, post-meter leakage and LNG follow deliveries or send-out, which are different physical drivers from field production. In the absence of appropriate observed monthly controls, those streams remain days-weighted or use a documented supported proxy. Refinery intake, wellhead production, flaring and venting retain distinct timing targets. The reference-year allocation does not imply observed changing geography across the extended annual series.

4.4. Air-pollutant source boundaries

The common source account supports admitted NMVOC and other pollutant factors for the specified oil/gas processes and flaring. Flaring combustion and venting are distinct, and factor coverage does not establish that every gas balance discrepancy is methane leakage. NH_3 and unresolved venting, post-meter or other pollutant categories remain notation gaps where no supported factor exists. The air ensemble reuses native fugitive activity realisations; an industrial gas-combustion draw deducts the same post-meter leakage quantity before applying a combustion factor.

4.5. Process-matched allocation

Production and gathering follow provincial oil and gas production. Transmission and distribution follow measured pipeline length, post-meter industrial emissions follow gas consumption, refineries follow facility output, and LNG terminals remain assigned to Sindh. Each allocation reconciles exactly to its national control. Facility and network geometry is insufficient for a defensible district grid, so the spatial product stops at process-matched province and point layers.

The implementation tests unit exponents, source-row and segment closure, activity-year anchors, pipeline and gas-flow identities, structural-model enumeration, uncertainty ordering, observation coverage, official-inventory reproduction, provincial reconciliation and manuscript-output agreement. Detailed assertions and generated products are documented in the supplement and reproducibility appendix.

5. Uncertainty and physical controls

5.1. Parametric uncertainty

The legacy experiment samples each published 2006 factor from a bounded log-uniform distribution and reports the median, P2.5, P97.5 and full methodological bounds. Rows in the same physical stream share their range quantile.

For the 2019 inventory, published IPCC limits are interpreted as bounded expert ranges. A split-triangular distribution places the reported value at the mode and median while preserving the stated lower and upper limits. One production-volume deviate is shared across gas production, gathering and processing; oil production and its transport fallback share the oil deviate; and gases within a physical factor group share a quantile. Additional processing and coverage residuals remain independent. The simulation uses 10 000 iterations and seed 42. The exact transformation, correlation groups and material inputs are reported in the supplement.

Input importance is measured by Spearman rank correlation with the national total. Normalized squared correlations provide a screening measure of the parameters controlling the conditional interval. A distribution-shape sensitivity reruns the reference model with bounded uniform and bounded log-uniform inputs.

5.2. Structural configurations

The factorial experiment crosses upstream practice (higher/lower), network practice (higher/lower), T&D activity basis (pipeline length/throughput), post-meter industrial coverage (fertilizer feedstock excluded, the reference/feedstock included/whole fertilizer class excluded/half/excluded), processing activity (gross production/marketable supply/excluded) and sour-gas share (32%/16%/excluded). Removing invalid combinations leaves 280 deterministic, unweighted configurations. No probability weights are assigned. We report one envelope for models retaining every public-data segment and a wider envelope that permits proxy-supported processing and post-meter coverage to be excluded. A separate named *central* configuration takes every stream with published high/low practice bounds at their geometric mean; it is priced, never selected, and is not a factorial leg because the practice bounds are readings of one control, not coverage choices.

Parametric intervals are calculated separately for five interpretable configurations: reference, central, lower practices, no post-meter industrial/power, and the minimum-coverage combined case. This keeps uncertainty in measured activities and factors conditional on the selected infrastructure model.

5.3. Gas-flow balance and UFG diagnostic

For each matched year, the physical account separates upstream and downstream identities:

$$F_{gross} = F_{indigenous,marketable} + R_{upstream}, \tag{2}$$

$$F_{indigenous,marketable} + F_{RLNG} = F_{use} + F_{UFG} + R_{downstream}. \tag{3}$$

The residuals preserve differences among published boundaries. The upstream disposition table also records reinjection, own use, processing shrinkage, venting and flaring where public quantities are available. Calendar-2024 VIIRS flare volume is shown as a volume-equivalent memo against the adjacent gross-to-marketable difference.

UFG includes physical losses, theft, metering error and accounting effects. It is therefore used only as a plausibility bound. If fraction q were physical loss, the available methane would be

$$M_{\mathrm{CH_4}} = qV_{\mathrm{UFG}}\rho_{gas}x_{\mathrm{CH_4}}. \tag{4}$$

Reference and lower-practice transmission-plus-distribution methane are expressed as the implied fraction q; post-meter emissions are outside this utility balance.

6. External evaluation

6.1. Official-inventory reconstruction

Pakistan's 2022 Common Reporting Table (CRT) is evaluated first in native gases. Common-metric comparisons use CO_2=1, fossil CH_4=30 and N_2O=265; the submitted total, which implies CH_4=28, is also retained. A matched-year waterfall begins with the common-metric CRT, removes separately reported venting and flaring already included in the 2019 Tier 1 factors, replaces matched oil and gas categories, and adds public-data segments absent from the CRT. Every published activity–factor product, native-gas subtotal and waterfall step is reproduced programmatically. Adjacent FY2021–22 and FY2022–23 inventories bracket calendar 2022.

6.2. Satellite and field evaluation

World Bank flare volumes and UNEP MARS methane sources are joined to provincial boundaries and to the 73 fields in the official gas-supply table. Each MARS source is assigned through its nearest named flare; official fields without observations remain as zero-observation controls. The evaluation reports marketable-supply coverage, Spearman and Kendall rank association, top-10 overlap, and national and within-province permutation tests for flare volume, MARS source count and plume count. A match audit records distance, operator, canonical field and confidence for every MARS source.

These observations evaluate upstream production and processing geography. Transmission, distribution and post-meter sources require utility or facility measurements. Flare conversion and MARS fluxes remain observational memos and do not enter the segment inventory.

6.3. Evidence roles

Table 2 assigns each comparison a specific task. This separation prevents an activity control, official account or intermittent observation from being treated as a universal benchmark.

7. Results

7.1. Segment completion matters more than factor revision

The legacy 2006 calculation is 3.805 Mt CO_2e for FY2019–20 and 3.317 Mt on FY2023–24 production. Applying 2019 factors to the same FY2023–24 production raises the estimate to 4.485 Mt. Adding gathering and processing gives 10.095 Mt; the full public-data ledger gives 23.499 Mt—high in the sector's 9.601 Mt to 24.694 Mt structural envelope, whose top keeps fertilizer feedstock inside the post-meter leak base (Figure 2). The factor-framework revision is therefore smaller than the effect of representing additional supply-chain segments.

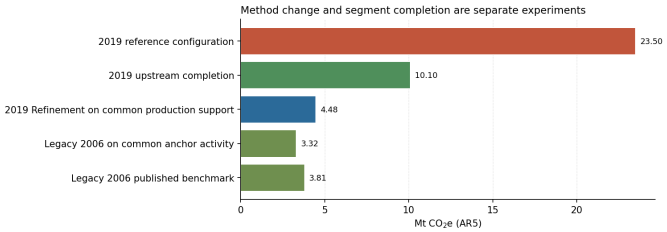


Figure 2: Method-resolution ladder. Common production support isolates the factor-framework effect before the account expands to additional segments.

7.2. Distribution and post-meter assumptions determine the range

The reference inventory contains 752.761 Gg CH_4, 915.619 Gg CO_2 and 3.483 t N_2O (Table 3). Distribution is the largest segment (7.069 Mt), followed by gas production (4.011 Mt), post-meter industrial and power use (3.420 Mt) and gathering (3.113 Mt).

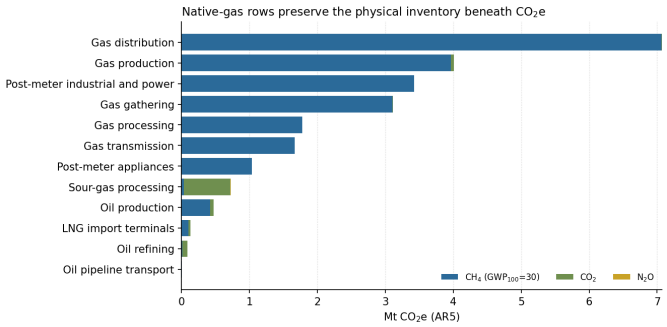


Figure 3: Segment-level contributions by native gas.

The lower distribution factor reduces the national total by 5.679 Mt. Its selection requires more than half plastic pipe and active LDAR; the publicly quantified 8200 km of SSGCL polythene covers only 4.2 % of the national network. Halving or excluding the transferred post-meter industrial/power factor reduces the total by 1.710 or 3.420 Mt, respectively.

The named lower-practice configuration has a deterministic value of 14.894 Mt. A synthesized *central* member—every stream with published high/low practice bounds at their geometric mean, the retired 2006 method's midpoint mechanism carried into the segment engine—prices 17.678 Mt, with its own conditional interval of 15.798 Mt to 21.309 Mt; it is a named configuration, never the selection, because no measurement locates Pakistani practice inside the bounds. The reference withholds the post-meter factor from the fertilizer feedstock—gas chemically consumed in the reformers, not burned—and keeps the plants' process fuel in the base. Putting the feedstock back inside the base, the membership of earlier releases and one this paper does not defend, prices 24.694 Mt (the post-meter feedstock scope axis, 1.195 Mt) and is the top of the structural envelope; taking the whole fertilizer class out prices 23.264 Mt (−0.236 Mt). Models retaining all public-data segments span 12.183 Mt to 24.694 Mt across the 192 segment-retaining members of the 280-configuration factorial.

Evidence	Question	Scale	Role
Gas-flow ledger	Are published supply and use controls compatible?	annual national	physical closure
UFG envelope	Is modelled network methane physically plausible?	annual utility	diagnostic bound
Pakistan CRT	Why does official reporting differ?	matched year/category	audited comparator
World Bank flares	Where is persistent combustion activity located?	field/site	spatial evaluation
UNEP MARS	Where are large detected methane sources?	event/field	spatial evaluation
Annual infrastructure controls	Does the reference result persist through time?	annual segment	temporal robustness

Table 2: Purpose and scale of the independent controls and comparisons.

Segment	CH_4 (Gg)	CO_2 (Gg)	N_2O (t)	Mt CO_2e	Share (%)	Activity
Gas distribution	235.439	6.037	0.000	7.069	30.1	official
Gas production	132.146	46.849	0.808	4.011	17.1	official
Post-meter industrial and power	113.976	1.140	0.000	3.420	14.6	official
Gas gathering	103.390	11.308	0.194	3.113	13.2	proxy
Gas processing	59.126	3.877	0.042	1.778	7.6	proxy
Gas transmission	55.459	3.787	0.000	1.668	7.1	official
Post-meter appliances	34.395	0.344	0.000	1.032	4.4	official
Sour-gas processing	1.034	688.579	0.558	0.720	3.1	IPCC default
Oil production	14.076	50.887	0.780	0.473	2.0	official
LNG import terminals	3.320	29.374	0.000	0.129	0.5	official
Oil refining	0.377	73.435	1.101	0.085	0.4	proxy
Oil pipeline transport	0.022	0.002	0.000	0.001	0.0	proxy
Total	**752.761**	**915.619**	**3.483**	**23.499**	**100.0**	

Table 3: Reference FY2023–24 inventory in native gases and AR5 CO_2 equivalent. Activity status distinguishes official controls from proxy-supported rows. The reference total sits high in the sector's 9.601–24.694 Mt structural envelope, whose top keeps fertilizer feedstock inside the post-meter leak base.

Permitting proxy-supported processing and post-meter coverage to be excluded expands the envelope to 9.601 Mt to 24.694 Mt. The current common conditional ensemble reruns the native bounded source/activity and factor model with shared physical identities. For FY2023–24 its reference 2.5th–97.5th percentiles are 21.3 Mt to 30.3 Mt CO_2e on AR5. Annual trend intervals use paired realisations; missing categories and structural configurations remain outside these percentiles. The standalone scenario diagnostic in Figure 4 retains its own declared settings and is not substituted for the current annual conditional product.

The distribution and post-meter factor groups contribute 73.2 % and 17.2 % of normalized squared rank sensitivity, respectively. All remaining groups individually contribute less than 2.5 %. Alternative bounded distribution shapes widen or shift the reference interval but leave these two measurement priorities unchanged (Supplementary Table S6).

7.3. Network growth offsets declining upstream activity

The public-data reconstruction ranges from 24.384 Mt in FY2015–16 to a peak of 26.270 Mt in FY2018–19 and 23.499 Mt in FY2023–24 (each year's figure the reference configuration, feedstock outside the leak base in every year). Its composition changes more clearly than its total. Upstream and processing emissions fall from 13.165 to 10.181 Mt (−22.7%), while network and LNG emissions rise from 6.698 to 8.866 Mt (+32.4%). Post-meter emissions remain nearly constant at 4.521 and 4.453 Mt. Growth in pipeline length and customer connections therefore offsets part of the decline in domestic production (Figure 6, right).

7.4. Physical balances and official reporting constrain magnitude

Gross production exceeds indigenous marketable supply by 11.4 %– 15.6 % across FY2021–22 to FY2023–24. The downstream balance is tighter: indigenous marketable supply plus RLNG closes to reported supply, and the residual after categorized consumption and UFG ranges from −2.6 % to a small positive residual over the reported annual series. These are source-account discrepancies, not measured atmospheric methane losses.

Reference transmission-plus-distribution emissions are 290.899 Gg CH_4, equivalent to 23.8 % of reported UFG methane. Lower-practice estimates require 6.1 %; higher and lower throughput cases require 18.8 % and 5.8 %. Both practice classes fit within the utility mass balance but imply sharply different physical-loss shares (Figure 5).

Pakistan's 2022 CRT reports 4.433 Mt for 1.B.2, or 4.749 Mt on the common GWP basis. The adjacent FY2021–22 and FY2022–23 reconstructions are 5.30 and 5.00 times this value. Fiscal-year choice changes the reconstruction by about 5.7 %.

Every CRT activity–factor row reproduces its reported native-gas result. The matched waterfall then closes at 25.187 Mt after harmonizing inclusive venting/flaring treatment, replacing the oil and gas segment calculations, and adding LNG and post-meter sources. The largest increments occur in gas production/gathering, processing, distribution and post-meter activity (Figure 6, left).

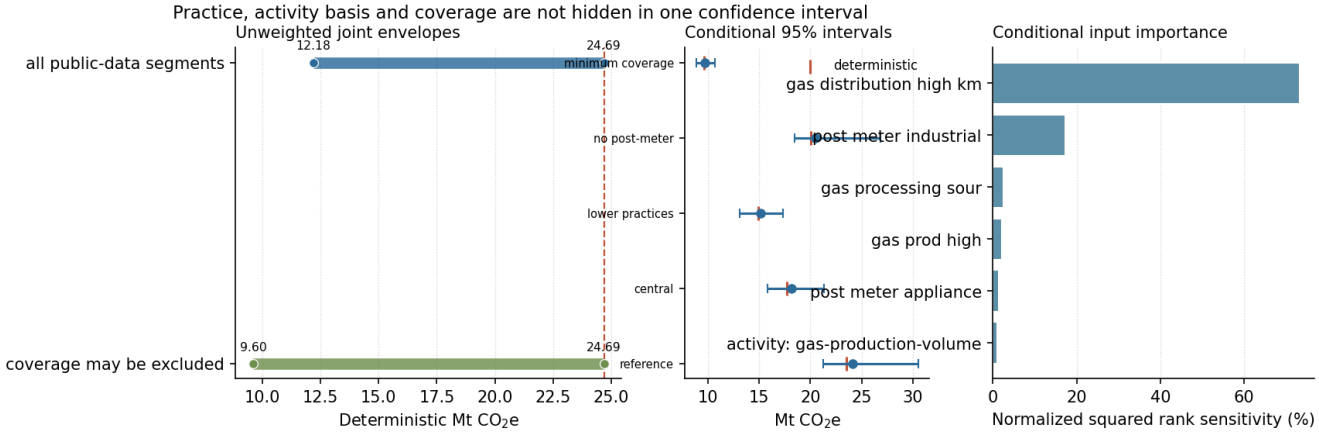


Figure 4: Structural envelopes, conditional parametric intervals and reference-configuration input importance.

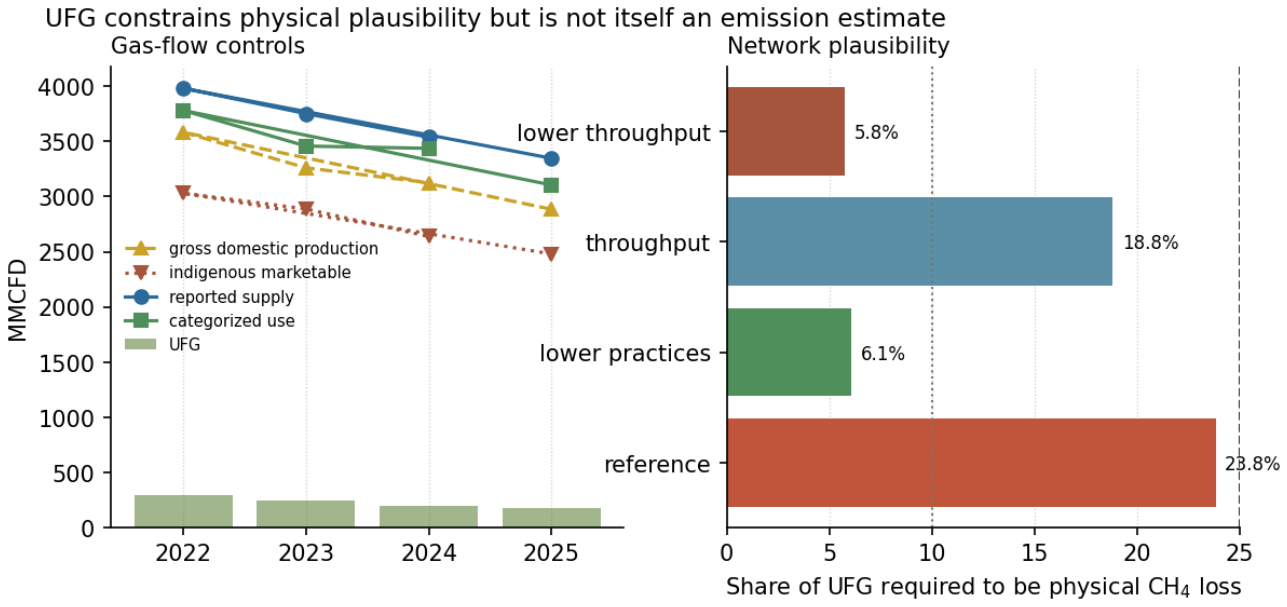


Figure 5: Reported gas-flow controls (left) and the UFG share corresponding to modelled pre-meter network methane (right).

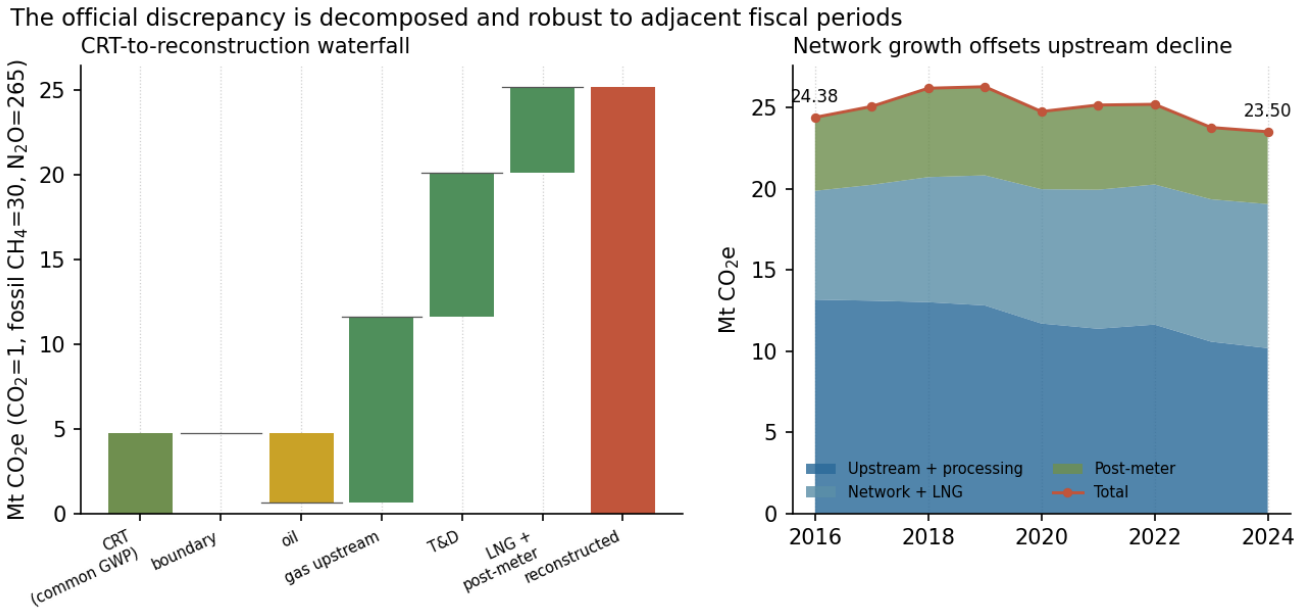


Figure 6: CRT-to-reconstruction waterfall on a common GWP basis (left) and the segment-resolved FY2015–16 to FY2023–24 reference-span reconstruction (right); the common annual core also includes FY2024–25.

7.5. Production locates sources but weakly predicts observed intensity

World Bank flare volume declines from 0.336 billion m^3 in 2012 to 0.130 in 2024. The MARS snapshot contains 20 oil-and-gas sources and 174 plumes; 19 sources occur within 10 km of a World Bank flare. Matched detections occurred near fields accounting for 71.1 % of reported supply—a statement about where the instruments matched, not about coverage: without sensor-opportunity, cloud and detection-limit masks, a field with no matched event is unobserved, not observed-clean.

Across all 73 official fields, production has Spearman ρ = 0.230 with VIIRS flare volume and ρ = 0.382 with MARS source and plume counts. Four fields overlap each observation's production top-10. National and within-province permutations place the MARS rank statistics at $p \leq 0.0013$ and the corresponding top-field overlaps at p = 0.0065–0.0374. The VIIRS rank result is weaker, while its top-10 overlap is less consistent with chance. Production therefore identifies broad upstream geography but explains little of the observed intensity variation (Figure 7).

The process-matched provincial inventory is 9.99 Mt in Sindh, 7.77 in Punjab, 2.86 in Khyber Pakhtunkhwa, 2.79 in Balochistan and 0.09 in Other.

8. Discussion

8.1. Segment completion changes national accounting

The common-support experiment identifies a modest increase from the 2006 to the 2019 production factors. The much larger change appears when gathering, processing, network and post-meter sources enter the account. For data-scarce countries, applying a newer production factor while retaining a production-only boundary would therefore capture only a small part of the Refinement's effect.

The segment-resolved reconstruction reinforces this interpretation. Domestic gas production declines, but distribution length and customer connections grow. The resulting shift from upstream to network emissions leaves the national total comparatively stable. A production-scaled time series would miss that structural change and direct mitigation attention toward a shrinking share of the system.

8.2. Two infrastructure questions define the national range

Distribution practice and post-meter industrial leakage determine most of both the structural range and the reference-configuration sensitivity. National pipe material, pressure, age and annual LDAR coverage would establish whether Pakistan meets the lower distribution-factor criteria. Public utility reports confirm ongoing leak surveys and rehabilitation, but the quantified material evidence covers only a small part of the network.

The industrial/power factor presents a different problem. Its IPCC lineage and unit conversion are explicit, and national activity is available by end-use class, but its empirical technology population comes from an older global default. The full, half-factor and exclusion cases show the consequence of that transfer. Measurements at power stations, general-industry sites and captive-power facilities would address the largest end-use components directly.

The reference configuration provides a reproducible application of the public evidence. The factorial envelopes show how alternative practice, activity-basis and coverage choices affect that result, while the conditional intervals quantify uncertainty within named configurations. Together they identify the obser-

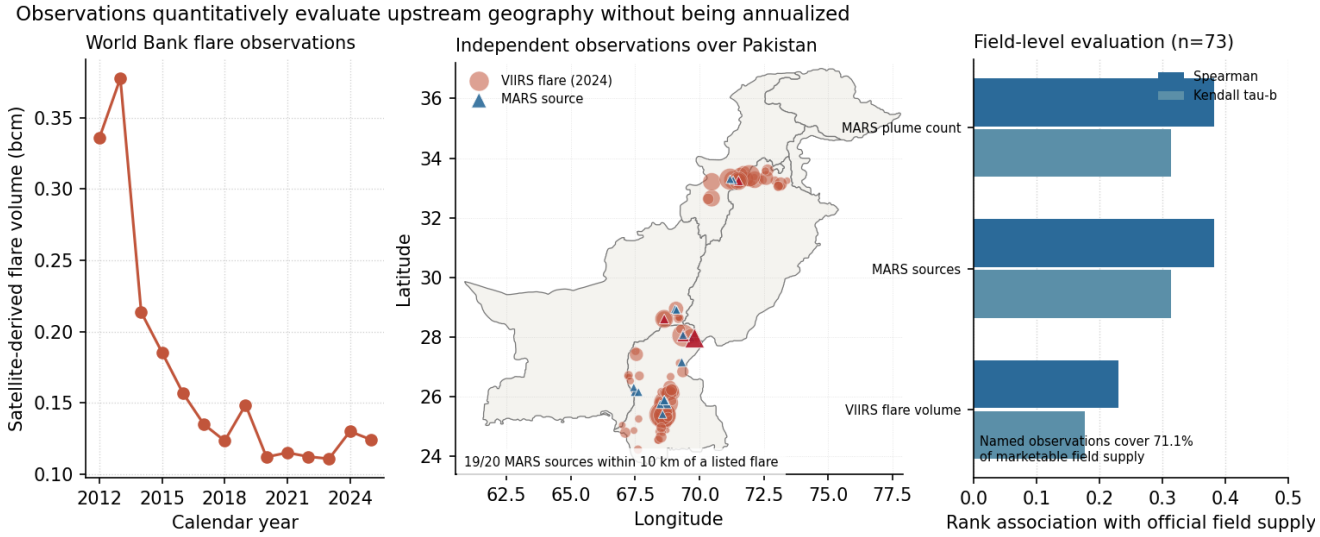


Figure 7: Annual flare volume, mapped field/flare/MARS observations and field-level ranking diagnostics.

vations capable of reducing uncertainty instead of compressing all model choices into one percentage.

8.3. Utility and official accounts identify reconciliation priorities

The gas-flow ledger establishes a coherent downstream system from indigenous marketable supply and RLNG to categorized use and UFG. Its unresolved upstream residual highlights a separate data need: plant throughput, reinjection, own use, processing shrinkage and operator venting/flaring. Those quantities would also improve the processing activity basis.

The UFG comparison translates network estimates into a quantity utilities can test. The reference and lower-practice configurations require markedly different physical shares of the reported balance, creating a direct role for utility loss audits that separate leakage from theft, metering and billing effects.

The CRT reconstruction locates the official difference in source representation. Common GWP conversion and adjacent fiscal years remove metric and period alignment as major explanations; the public workbook itself is arithmetically coherent. The waterfall instead points to gas production/gathering activity, processing, distribution, post-meter coverage and the oil–gas allocation. Reconciliation should therefore begin with the underlying activity bridge and category boundaries.

8.4. Satellite observations reveal heterogeneous upstream emissions

The field analysis supports the broad geography of the upstream inventory while challenging a simple production-intensity relationship. Observations cover most reported marketable supply, and MARS source locations are non-random with respect to producing fields. Their modest rank association with production is consistent with equipment, operating practice and intermittent events controlling emissions at individual sites.

This finding has two implications. First, production remains useful for allocating an aggregate upstream estimate when facility activity is incomplete. Second, production cannot substitute for facility observations when identifying high emitters. Repeated plume coverage, plant and compressor inventories, and operator-matched field data would turn the present geographic test into a stronger facility-level evaluation.

8.5. Implications for data-scarce inventories

The source-neutral ledger makes future improvements local. Well and field records can replace national production proxies; plant throughput can resolve processing and disposition; utility material and pressure classes can select network factors; and facility measurements can replace transferred post-meter factors. The evidence ranking indicates that national distribution technology, post-meter measurements and upstream disposition should precede refinements to minor oil-transport or LNG rows.

More generally, the study shows why factor selection, boundary completion and evaluation should be treated as separate tasks. A newer factor framework improves an inventory only when its physical segments are represented, while independent controls are most informative when matched to the processes they can observe.

9. Limitations

Pakistan-specific equipment and leakage measurements remain sparse. National pipe material, pressure class, network age, annual LDAR coverage, compressor type and seal configuration were unavailable. The post-meter industrial/power factor is transferred from the IPCC default, and processing uses produced or marketable gas in the absence of plant throughput. These inputs control the largest structural differences.

The post-meter factor also rides gas the combustion sectors meter: the national gas ledger (`outputs/national_gas_`

closure.csv) joins OGRA Table 3.31's end-use partition to each consuming module's live quantity and quantifies the leaked share of the burned-and-leaked classes' double-claim at about 1.0 % of their whole-gas mass. The combustion side books its gas rows *net* of that share, so no molecule is claimed both burned and leaked, and this sector's leak claim is unchanged. The gross delivered-gas booking is the joint national configuration *gross_delivered_gas_accounting* (0.288 Mt of combustion CO_2e restored onto the gas rows the consuming modules book net; outputs/national_gas_closure_variant.csv). That figure is the combustion side of the four burned streams and holds no fertilizer term. Fertilizer gas is the ledger's sharpest single class, 31 % of the six classes the leak base is drawn from, and it is carried as two rows. The chemical feedstock, 83.5 % of the class, is reformed and not burned: it sits outside every combustion module and outside the leak base. The process fuel is burned at the plants and stays inside it. A membership with the feedstock inside the base is published as the scenario *post_meter_including_feedstock* (24.694 Mt), the highest case of the structural envelope; it carries no probability and is not the reference.

The public gas-flow account does not resolve reinjection, own use, processing shrinkage or operator venting and flaring. Active and abandoned wells, exploration, workovers, pneumatic equipment, blowdowns, accidents and anomalous releases remain NE or embedded in inclusive factors according to the completeness matrix. A complete well-, facility- and component-level ledger would improve both coverage and technology classification.

Satellite evidence is spatially and temporally selective. VIIRS detects combustion radiance, MARS emphasizes large intermittent plumes, and only 8 of the 73 official fields receive matched MARS events. A zero in the matched record is therefore not an observed non-detection: it may be unobserved, cloud- or geometry-limited, below the detection threshold, or outside sensor coverage, and the permutation tests retain those zeros as rank evidence only—no field is read as observed-clean. Building the sensor-opportunity cube that would separate these cases is named future work. The field analysis consequently tests upstream geography; network and post-meter rows need utility, facility or urban atmospheric measurements.

The full infrastructure reconstruction begins in FY2015–16, and fiscal years only bracket the calendar-2022 CRT comparison. Structural envelopes cover the declared public-data choices but cannot enumerate every physical configuration. Conditional intervals also depend on the bounded input distributions documented in the supplement.

10. Conclusion

On common FY2023–24 production support, the legacy and refined methods give 3.317 and 4.485 Mt CO_2e. Expanding to the public-data 2019 segment ledger raises the conservative (default-practice) configuration to 23.499 Mt—high in its own 9.601–24.694 joint envelope, whose top is the same practice setting with fertilizer feedstock inside the post-meter leak base. The reference should be read as an upper-practice conservative configuration and never as an empirically centred best estimate: no probability attaches to any point in the envelope, and until Pakistani pipe-material, pressure, LDAR or post-meter measurements exist, the honest statement is the envelope with its selection rule, with the lowest joint configuration published beside it. The principal change is therefore supply-chain coverage rather than the production-factor revision alone.

Joint structural choices span 9.601 Mt to 24.694 Mt. Distribution technology and transferred post-meter industrial leakage determine most of this range and 90.4 % of the reference-configuration rank sensitivity. Utility balances show that the network estimates are physically feasible, while distinguishing sharply between higher- and lower-practice interpretations.

Both adjacent-fiscal-year bracket endpoints are more than five times Pakistan's 2022 CRT on a common GWP basis; the difference traces to activity, factor and boundary choices rather than arithmetic. Satellite observations confirm broad upstream source geography but show that production weakly predicts detected intensity. National pipe characteristics, post-meter measurements and plant gas disposition are therefore the highest-value next data. The implemented ledger can incorporate those records without another methodological redesign.

Code and data availability

The calculation engine, the activity data and every output quoted here are version-controlled in the inventory repository (https://github.com/PakAirQuality/pakistan-emissions-sectors) and reproduce from a single command. The sector is regenerated with python3 -m pkinv.sectors.fugitives.run_fugitives; Appendix A gives the full sequence. Every number printed in this paper is read from a generated CSV rather than transcribed, and a test asserts that the manuscript and the engine agree.

References

Alvarez, R.A., Zavala-Araiza, D., Lyon, D.R., Allen, D.T., Barkley, Z.R., Brandt, A.R., Davis, K.J., et al., 2018. Assessment of methane emissions from the U.S. oil and gas supply chain. Science 361, 186–188. doi:10.1126/science.aar7204.

Chen, Z., Jacob, D.J., Nesser, H., Sulprizio, M.P., Lorente, A., Tadić, J.M., Gautam, R., et al., 2023. National quantifications of methane emissions from fuel exploitation using high-resolution inversions of satellite observations. Nature Communications 14, 4948. doi:10.1038/s41467-023-40671-6.

Cusworth, D.H., Duren, R.M., Thorpe, A.K., Olson-Duvall, W., Heckler, J., Chapman, J.W., Eastwood, M.L., et al., 2021. Intermittency of large methane emitters in the permian basin. Environmental Science & Technology Letters 8, 567–573. doi:10.1021/acs.estlett.1c00173.

Elvidge, C.D., Zhizhin, M., Baugh, K., Hsu, F.C., Ghosh, T., 2016. VIIRS Nightfire: satellite pyrometry at night — global flared-gas volume estimates. Earth Observation Group, Colorado School of Mines. The instrument and method behind the World Bank individual-flare workbook this inventory evaluates against; 10.3390/rs8010017.

Finance Division, Government of Pakistan, 2026. Pakistan Economic Survey 2025–26. Technical Report. Ministry of Finance. Islamabad. URL: https://www.finance.gov.pk/survey_2026.html.

Government of Pakistan, 2024. Pakistan's First Biennial Transparency Report to the UNFCCC. Technical Report. Ministry of Climate Change and Environmental Coordination. URL: https://unfccc.int/documents/645241.

Government of Pakistan, 2026. Common reporting tables for the electronic reporting of pakistan's national greenhouse gas inventory. Submission to the UNFCCC. URL: https://unfccc.int/documents/655653.

IPCC, 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2: Energy, Chapter 4: Fugitive Emissions. Technical Report. Institute for Global Environmental Strategies, Hayama, Japan. URL: https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html.

IPCC, 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Technical Report. IPCC, Switzerland.

Lauvaux, T., Giron, C., Mazzolini, M., d'Aspremont, A., Duren, R., Cusworth, D., Shindell, D., Ciais, P., 2022. Global assessment of oil and gas methane ultra-emitters. Science 375, 557–561. doi:10.1126/science.abj4351.

National Fertilizer Development Centre, 2025. Fertilizer Review 2024–25. Technical Report. Planning Commission, Government of Pakistan. Islamabad.

Oil and Gas Regulatory Authority, 2023. State of the Regulated Petroleum Industry 2021–22. Technical Report. Government of Pakistan. URL: https://www.ogra.org.pk/petroleum-industry-reports.

Oil and Gas Regulatory Authority, 2024a. State of the Regulated Petroleum Industry 2022–23. Technical Report. Government of Pakistan. URL: https://www.ogra.org.pk/petroleum-industry-reports.

Oil and Gas Regulatory Authority, 2024b. State of the Regulated Petroleum Industry 2023–24. Technical Report. Government of Pakistan. Islamabad. URL: https://www.ogra.org.pk/annual-reports.

Oil and Gas Regulatory Authority, 2026. Petroleum industry reports archive. Official report archive. URL: https://www.ogra.org.pk/ogra-annual-reports. accessed 7 August 2026.

Runfola, D., Anderson, A., Baier, H., et al., 2020. geoBoundaries: A global database of political administrative boundaries. PLOS ONE 15, e0231866. doi:10.1371/journal.pone.0231866.

Scarpelli, T.R., Jacob, D.J., Maasakkers, J.D., Sulprizio, M.P., Sheng, J.X., Rose, K., Romeo, L., et al., 2020. A global gridded ($0.1^\circ \times 0.1^\circ$) inventory of methane emissions from oil, gas, and coal exploitation based on national reports to the UNFCCC. Earth System Science Data 12, 563–575. doi:10.5194/essd-12-563-2020.

Scarpelli, T.R., Jacob, D.J., Moran, M., Reuland, F., Gordon, D., Maasakkers, J.D., Sulprizio, M.P., et al., 2022. Updated global fuel exploitation inventory (GFEI) for methane emissions from the oil, gas, and coal sectors: Evaluation with inversions of atmospheric methane observations. Atmospheric Chemistry and Physics 22, 3235–3249. doi:10.5194/acp-22-3235-2022.

Sui Northern Gas Pipelines Limited, 2024. Annual Report 2024: Financial Year 2023–24. Technical Report. Sui Northern Gas Pipelines Limited. URL: https://www.sngpl.com.pk/download/AnnualReport-2024/SNGPLAR2024.pdf.

Sui Southern Gas Company Limited, 2024. Annual Report 2024. Technical Report. Sui Southern Gas Company Limited. URL: https://www.ssgc.com.pk/web/wp-content/uploads/pdfs/2024_annual_report.pdf.

United Nations Environment Programme, 2026. Methane alert and response system: Sources and plumes dataset. International Methane Emissions Observatory, Eye on Methane. URL: https://methanedata.unep.org/methane-alert-response-system. versioned download accessed 6 August 2026.

Veefkind, J.P., Aben, I., McMullan, K., Förster, H., de Vries, J., Otter, G., Claas, J., Eskes, H.J., de Haan, J.F., Kleipool, Q., et al., 2012. TROPOMI on the ESA Sentinel-5 precursor: A GMES mission for global observations of the atmospheric composition for climate, air quality and ozone layer applications. Remote Sensing of Environment 120, 70–83. doi:10.1016/j.rse.2011.09.027.

World Bank, 2026. Global gas flaring data: Flare volume estimates by individual flare location, 2012–2025. Global Flaring and Methane Reduction Partnership. URL: https://www.worldbank.org/en/programs/gasflaringreduction/global-flaring-data. versioned workbook accessed 6 August 2026.

Zavala-Araiza, D., Lyon, D.R., Alvarez, R.A., Davis, K.J., Harriss, R., Herndon, S.C., Karion, A., et al., 2015. Reconciling divergent estimates of oil and gas methane emissions. Proceedings of the National Academy of Sciences 112, 15597–15602. doi:10.1073/pnas.1522126112.

Appendix A. Reproduction and generated products

The inventory is regenerated from the repository root with:

```
pip install -e .
python3 -m pkinv.sectors.fugitives.run_fugitives
python3 -m pkinv.report.figures
python3 scripts/sync_figures.py
python3 -m unittest discover -s tests
```

The runner writes the evidence tables below. Each is named: the registered filenames use the FY2023–24 anchor suffix 2024, and stating the convention without stating the names left a reader unable to tell which file was which. Every quantity this paper prints from one of them is checked against it by the verification registry.

- `fugitives_segment_native_gases_2024.csv`: the segment ledger in native gases: CH_4, CO_2 and N_2O for each of 12 segments, with each segment's share of the sector.
- `fugitives_method_ladder_2024.csv`: the method ladder, ordered, each rung with its coverage and result.
- `fugitives_structural_factorial_2024.csv`: all 280 factorial configurations of the practice, basis and sour-fraction choices.
- `fugitives_structural_scenarios_2024.csv`: the named structural experiments, each with its activity basis and practice.
- `fugitives_conditional_intervals_2024.csv`: 5 conditional Monte Carlo intervals, one per configuration, beside that configuration's deterministic value.
- `fugitives_uncertainty_attribution_2024.csv`: rank attribution by input group, as normalised squared Spearman correlation.
- `fugitives_infrastructure_timeseries_2016_2024.csv`: the 9-year infrastructure reconstruction.
- `fugitives_infrastructure_segment_timeseries_2016_2024.csv`: the same years decomposed by segment.
- `fugitives_gas_flow_balance_2022_2024.csv`: the gas-flow balance: gross domestic production, indigenous marketable supply and the upstream statistical difference between them.
- `fugitives_ufg_network_compatibility_2024.csv`: the UFG test—the physical fraction each scenario would require, and whether it is compatible with a 10 % and a 25 % unaccounted-for-gas figure.
- `fugitives_post_meter_end_use_2024.csv`: post-meter throughput by end use, with the IPCC inclusion decision for each.
- `fugitives_post_meter_factor_lineage_2024.csv`: that factor's lineage from its published value to the registered one, rounding difference included.
- `fugitives_distribution_practice_evidence_2024.csv`: the evidence that classifies the distribution network, each item with its criterion, coverage and effect on the classification.
- `fugitives_observation_field_statistics_2024.csv`: field-level agreement with the satellite observations.
- `fugitives_observation_permutation_2024.csv`: the permutation tests that give that agreement an empirical p.
- `fugitives_data_priorities_2024.csv`: which input or decision would move the answer most, ranked by effect.
- `fugitives_benchmark_crosswalk.csv`: the external comparators on a common GWP basis, each with its independence from this inventory recorded.
- `fugitives_mars_summary.csv`: the MARS plume summary—sources, plumes, persistent and notified counts and the maximum observed flux.
- `fugitives_viirs_flare_sites.csv`: the flare-site register, with each site's volume per year.
- `fugitives_viirs_flaring_2012_2025.csv`: the national series of flared volume and active site count.

The sector's headline tables are the seven below. They were described in this appendix by function—"the process-matched provincial allocation", "the completeness matrix"—and named nowhere, which tells a reader that the evidence exists without telling them where it is.

- `fugitives_totals_2024.csv`: the registered sector result, by category and method, with the fiscal year, the AR5 total and the Monte Carlo 2.5th, 50th and 97.5th percentiles beside it.
- `fugitives_by_stream_2024.csv`: the stream ledger—category, subitem and gas in gigagrams, each row with its uncertainty, correlation group, fossil flag, activity year and period kind, evidence class, and temporal and spatial method.
- `fugitives_provincial_2024.csv`: the province-level AR5 total.
- `fugitives_monthly_2024.csv`: the twelve-month layer, each month's share with the emissions it implies in both common-GWP vintages, and the temporal method and basis that produced the shape.
- `fugitives_completeness_2024.csv`: one row per IPCC code and subsegment with its notation key, treatment, principal gap, materiality class and screening bracket.

- `fugitives_uncertainty_2024.csv`: per segment, the mean and the 2.5th, 50th and 97.5th percentiles with the observed minimum and maximum, and the method, scenario, distribution, iteration count and seed that generated them.
- `fugitives_observation_provincial_evaluation_2024.csv`: the provincial confrontation of the upstream inventory with the observations—flared volume, active and total flare sites, and MARS source and plume counts, persistent ones separately.

The runner also writes the reference component table, the auditable site matches, the CRT extraction, the common-GWP waterfall, the row reproduction and cell audit, the QA table and the run manifest. The legacy FY2019–20 calculation remains a labelled row rather than the registered sector result.

The run manifest records each input path and SHA-256 hash. Tests assert the registered year, matched-year activity conversion, IPCC factor identity, alternative activity bases, gas-flow identities, no duplicate segment–gas rows, uncapped bounded legacy sampling, shared uncertainty groups, UFG exclusion and compatibility, quantitative observation coverage, CRT row closure, provincial reconciliation and source-status notation.

Appendix B. Factor and activity audit

The machine-readable factor registry is `data/fugitives_ipcc2019_factors.csv`. Each row stores its 2019 Refinement table, activity unit, reported factor, lower and upper uncertainty, practice class and correlation group. The legacy factors remain in `data/emission_factors.csv` only for the method benchmark.

Reference activities are stored in `data/activity/fugitives/anchor_activity_2023_24.csv` and the matched-year extension in `data/activity/fugitives/anchor_activity_2021_23.csv`. Historical controls are in `data/activity/fugitives/infrastructure_history_2015_21.csv`; field, distribution-practice and post-meter audits have separate source tables in the same directory. Official quantities and fallbacks are distinguished in the `method_tier`, `status` and `note` fields. The factor engine accepts this generic schema; it does not read OGRA or satellite files directly.

The official-inventory audit exports the workbook filename and package hash, worksheet, cell or range, raw value, unit, formula where present, notation key and extraction transformation. This is intended to make the reported oil–gas allocation independently checkable rather than to infer unpublished institutional reasoning.

The complete uncertainty-input register, including distributions, bounds and shared deviates, is reproduced as Table S5 in the supplementary material. The machine-readable source remains `outputs/fugitives_uncertainty_inputs_2024.csv`.

Appendix C. Notation keys and import contract

Completeness follows the UNFCCC notation convention: *estimated* identifies ledger rows, IE identifies quantities included elsewhere in an inclusive factor, NE identifies categories for which evidence is not yet sufficient, and NO identifies activities documented as not occurring. Venting and flaring are IE in the reference Tier 1 production factors; the VIIRS result is consequently a memo, not an additional stream.

Table S4 in the supplementary material reproduces the complete notation-key and materiality matrix, including the activity evidence required to replace every NE entry. The observation join audit and its 10,000-draw permutation diagnostics are given in Tables S1–S3.

The permanent import contract for future PPIS, OGRA or operator records requires a source identifier, reporting period, IPCC segment, facility or network identifier, operator, location, activity value and unit, technology/practice class, source reference, imputation method and quality flag. Missing records remain explicit residuals. Facility observations may replace an aggregate activity row or evaluate its geography, but they cannot silently change the inventory boundary or force closure.

Appendix D. Legacy uncertainty experiment

For the 2006 benchmark, a published factor range $[L, U]$ is sampled directly as

$$EF = \exp\left[\ln L + q(\ln U - \ln L)\right], \qquad q \sim \mathcal{U}(0,1). \tag{D.1}$$

All gases in one physical stream share q. The geometric midpoint is therefore the median of the factor distribution, every draw remains within the published bounds, and no display or computational cap is applied. This bounded log-uniform experiment is retained only to diagnose the legacy method; it is not the uncertainty model used for the 2019 segment inventory.